## Purpose

Standard procedure for Root and Intermediate CA Renewal and Certificate Revocation Updates. Additionally, Revoking Procedures at the bottom.

**Table for Reference**

| PKI Component | Validity Period | Last Renewal | Expiration Date (Next Renewal) |
|---|---|---|---|
| Root Certificate | X Years | | |
| Root Certificate Revocation List | X Years | | |
| Intermediate Certificate | X Years | | |

### We should revoke a certificate if:

- Device is **stolen** (Attacker can pull private key from storage)
- Device is **compromised** (Attacker can borrow "identity" for any purpose)
- Device is **retired** (Attacker can scavenge private key and use a "legacy" identity)
- Certificate Authority is **stolen/compromised/retired** (Attacker mimics CA)

## Scope

- Renewing Certificate for Root Certificate Authority **(Every 10 years)**
- Renewing Certificate Revocation List for Root Certificate Authority **(Every 5 years)**
- Renewing Certificate for Intermediate Certificate Authority **(Every 5 years)**
- Revoking certificates and pushing out the new certificate revocation list **(Ad-hoc)**

# Procedure

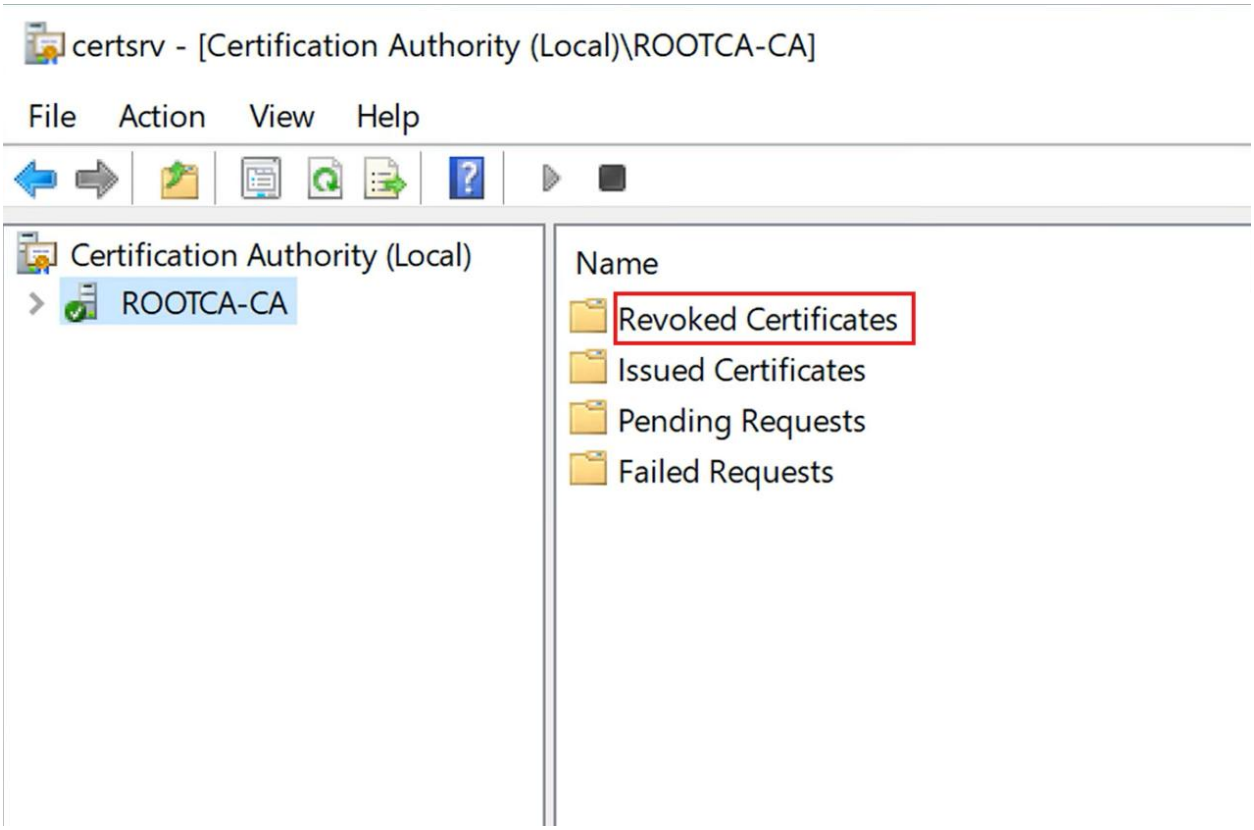
**Follow procedure based on the Maintenance Event (5 years / 10 years)**

## Routine Maintenance (Every 5 Years)

### (1) Renewing the Root Certificate Revocation List

**On Root CA Server**

1. In **Certificate Authority (certsrv.msc),**
    a. Right Click **Revoked Certificates >** Go to **All Task** > Click **Publish**

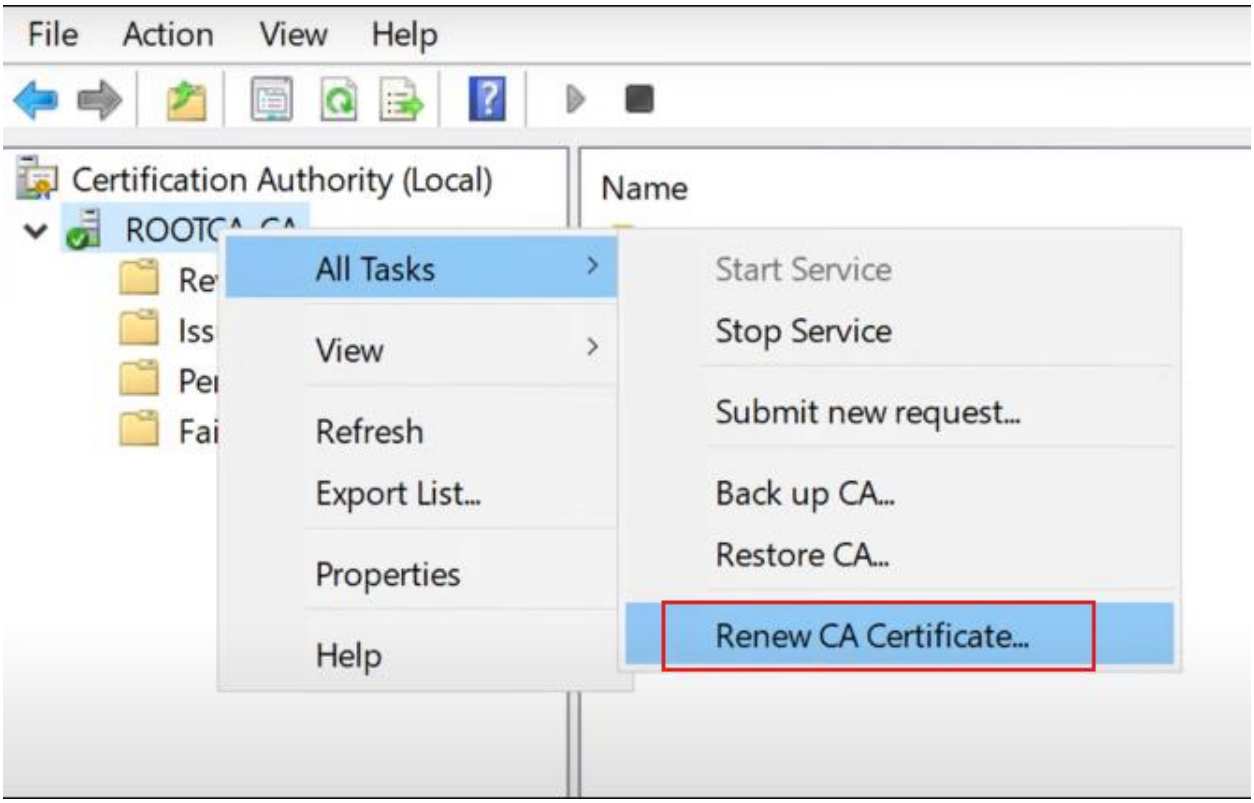



2. Transfer the new **.crl** file onto a secure media (found in **C:\Windows\System32\CertSrv\CertEnroll)**

## (2) Renewing the Intermediate CA Certificate / Transferring the Root CRL

**On Intermediate CA Server**

1. In **Certificate Authority (certsrv.msc),**
    a. Right Click **CA Name** > Go to **All Tasks** > Click **Renew CA Certificate**

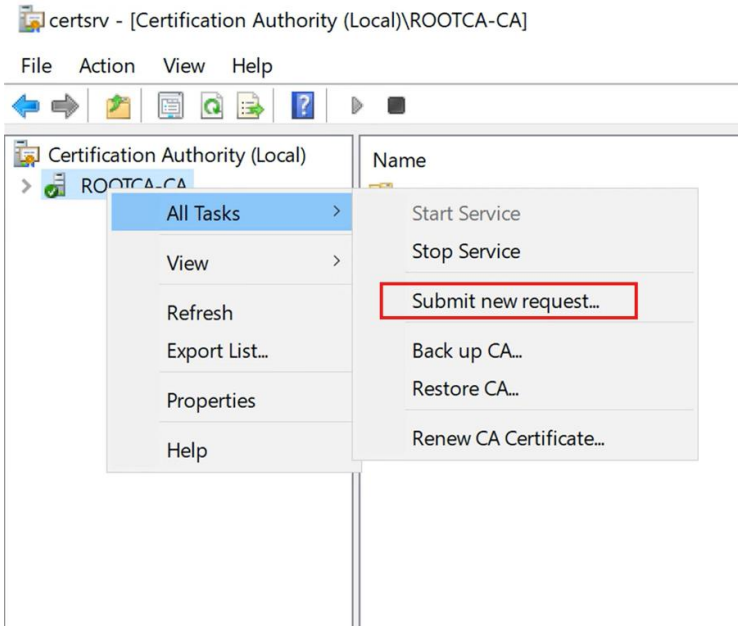



2. **In the pop-up box, do not generate a new private/public key**
3. Save the **.req** file onto a secure media
4. From secure media, transfer the lastest root **.crl** file into C**:\Windows\System32\CertSrv\CertEnroll**
5. Right-click the **.crl** file and select **Install**.
6. Set the **Certificate Store** as **Intermediate Certification Authorities**, finish wizard with defaults
7. In **C:\inetpub\wwwroot\CertEnroll**, replace the old root **.crl** file with the new file. **The name must be the same.**

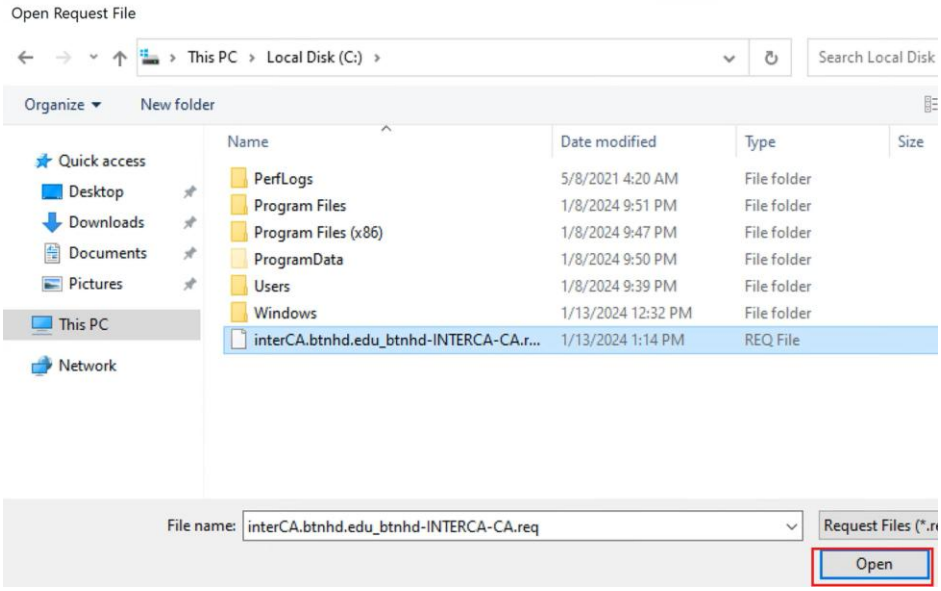
## (3) Signing the Intermediate CSR / Installing the Signed CSR

**On Root CA Server**

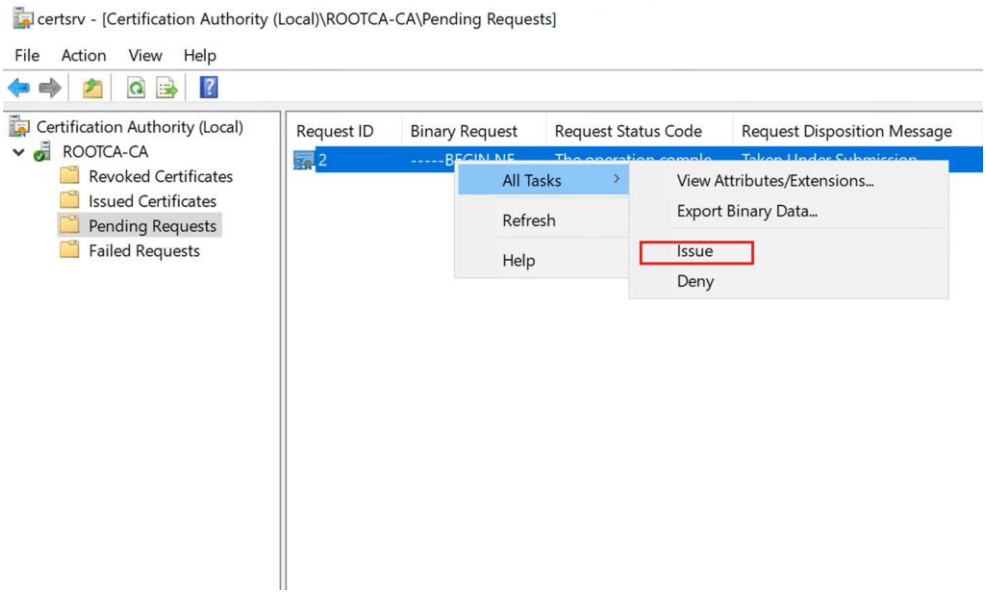
1. Mount secure media containing the **.req** file
2. Copy the latest **.req** file onto **C:**
3. In **Certificate Authority**, right-click the **CA Name** > go to **All Tasks** > click **Submit new request...**



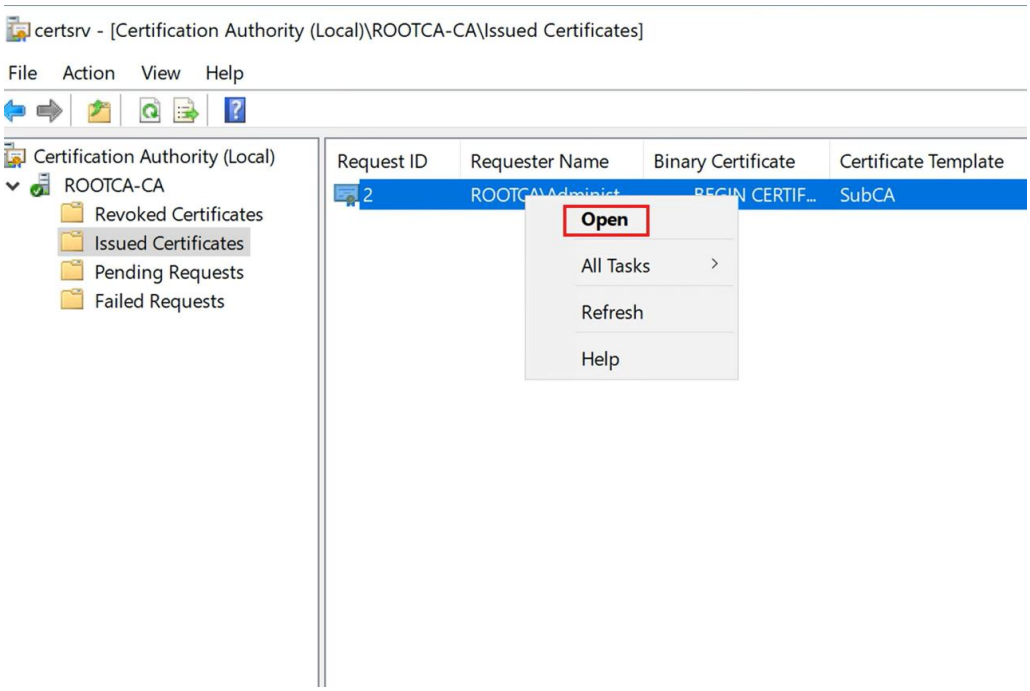

4. In the **Open Request File** window, open the latest imported **.req** file



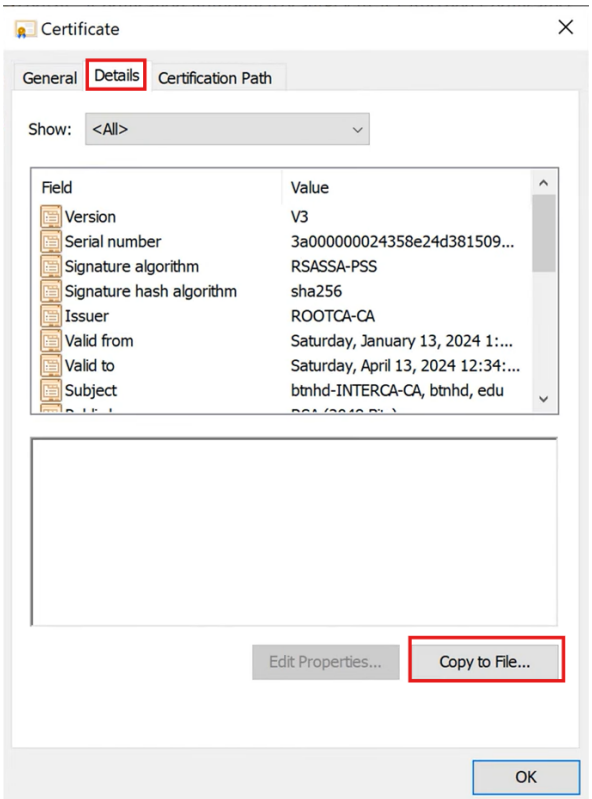
5. In **Certificate Authority**, go to **Pending Requests** > **right-click** the latest request > go to **All Tasks** > and click **Issue**



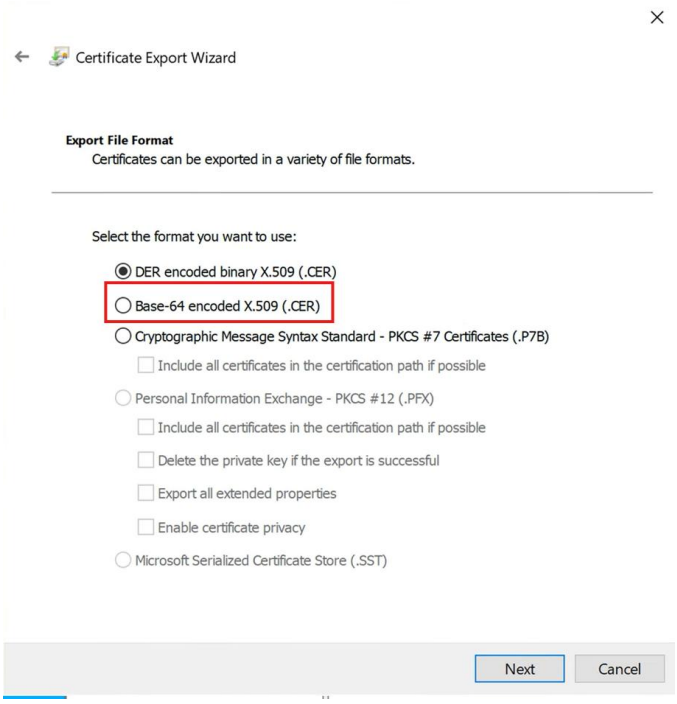

6. Go to **Issued Certificates > right-click** the latest certificate, and click **Open**



7. In the **Certificate** window, go to **Details** tab and **click Copy to File…**



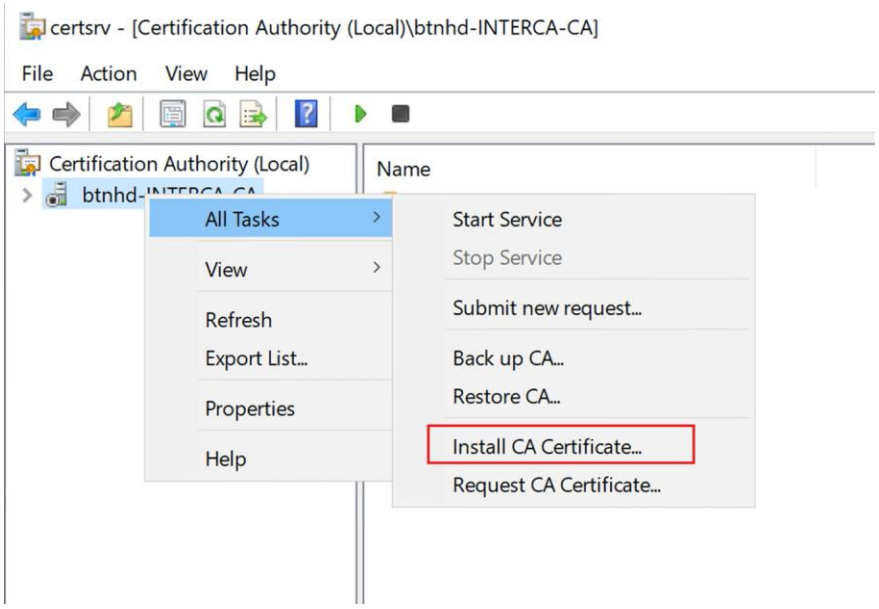

8. In **Certificate Export Wizard**, set the format to be **Base-64 encoded X.509 (.CER)** (this format should be readable for all SCADA device)



9. Continue with defaults, **Save** file name as **[domain_IntermediateCAName_CA].crt**
10. Transfer the latest Intermediate CA **.crt** file onto secure media

**On Intermediate CA Server**

1. Mount secure media containing latest Intermediate CA **.cer/.crt** file
2. In **Certificate Authority**, right-click the **CA Name** > go to **All Tasks** > select **Install CA Certificate**




3. Open the latest Intermediate CA **.cer/.crt** file
4. In **C:\inetpub\wwwroot\CertEnroll**,
    a. Replace the old Intermediate CA **.cer/.crt** file with the latest (name must be the same)

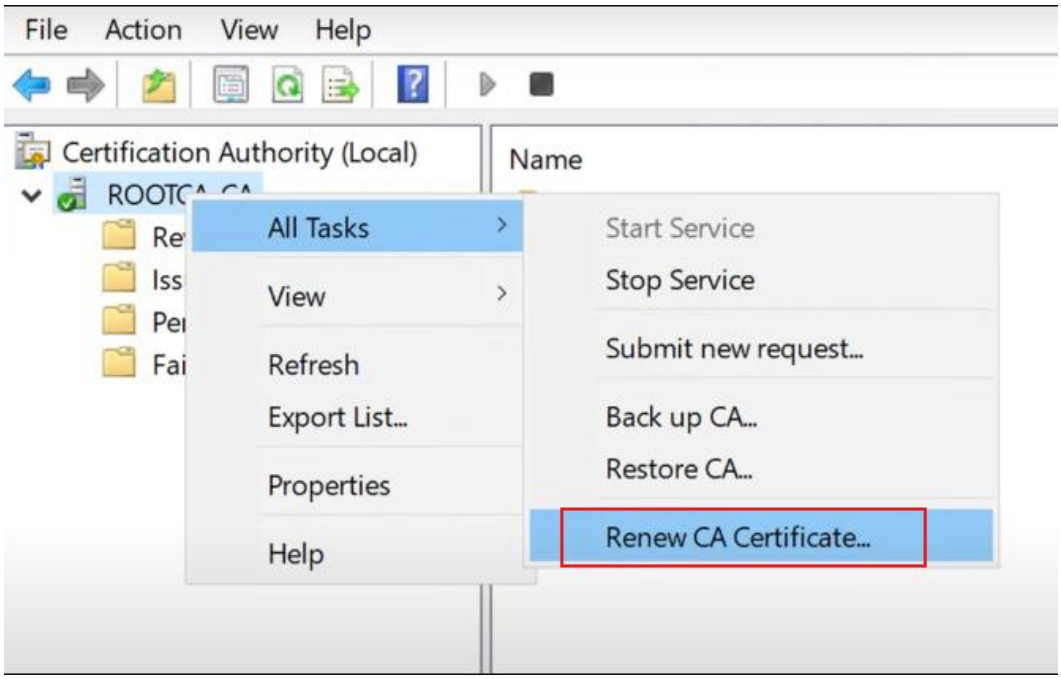
**Restart the Certificate Services service**

## Complete Maintenance (Every 10 Years)

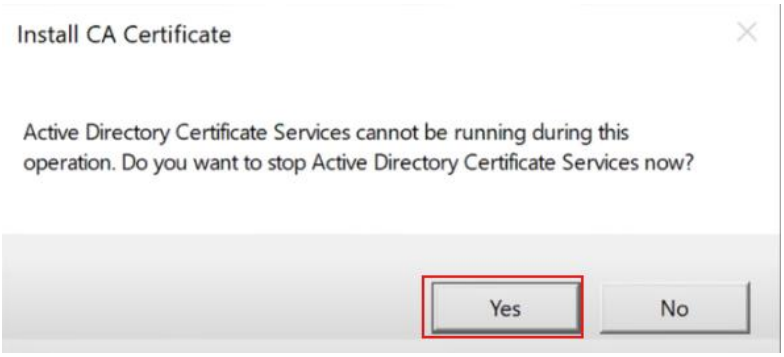
### (1) Renewing the Root CA Certificate

**On Root CA Server**

1. In **Certificate Authority**, right-click **CA Name** > go to **All Tasks** > select **Renew CA Certificate...**




2. Stop Active Directory Certificate Services to continue

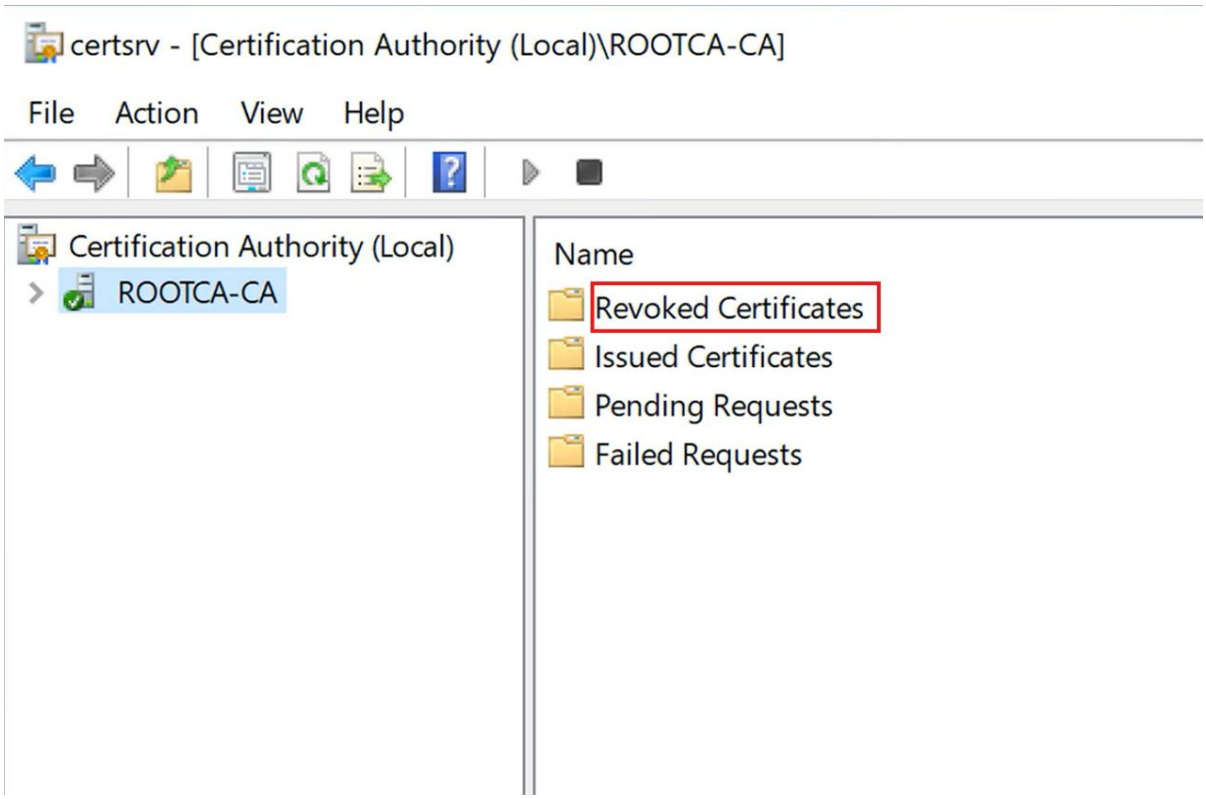



3. **In the pop-up box, do not generate a new private/public key**
4. Under **C:\Windows\System32\CertSrv\CertEnroll**, transfer the latest root **.cer/.crt** file onto secure media
   a. **Must replace all instances of old root certificate with latest**

## (2) Renewing the Root Certificate Revocation List

**On Root CA Server**

1. In **Certificate Authority (certsrv.msc),**
    a. Right Click **Revoked Certificates >** Go to **All Task** > Click **Publish**



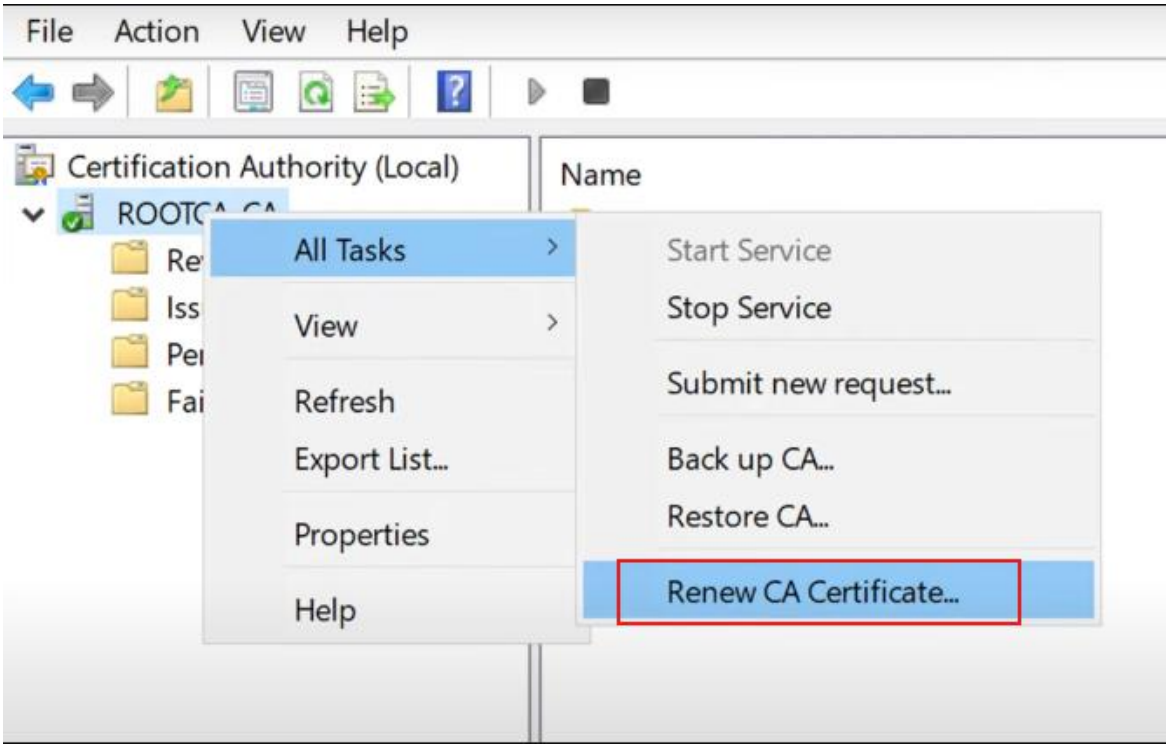

2. Transfer the new **.crl** file onto a secure media (found in **C:\Windows\System32\CertSrv\CertEnroll)**

## (3) Transferring the Root Cert and CRL / Renewing Intermediate CA Certificate

**On Intermediate CA Server**

1. From secure media, transfer the lastest root **.crl** and **.crt/.cer** files into C**:\Windows\System32\CertSrv\CertEnroll**
2. Right-click the root **.crt/.cer** file and select **Install**
3. Set the **Certificate Store** as **Trusted Root Certification Authorities**, finish wizard with defaults
4. Right-click the **.crl** file and select **Install**.

5. Set the **Certificate Store** as **Intermediate Certification Authorities**, finish wizard with defaults
6. In **C:\inetpub\wwwroot\CertEnroll**, replace the old root **.crl** and **.crt/.cer** files with the new files. **The name must be the same or else logic filepath fails.**
7. In **Certificate Authority (certsrv.msc),**
    a. Right Click **CA Name** > Go to **All Tasks** > Click **Renew CA Certificate**

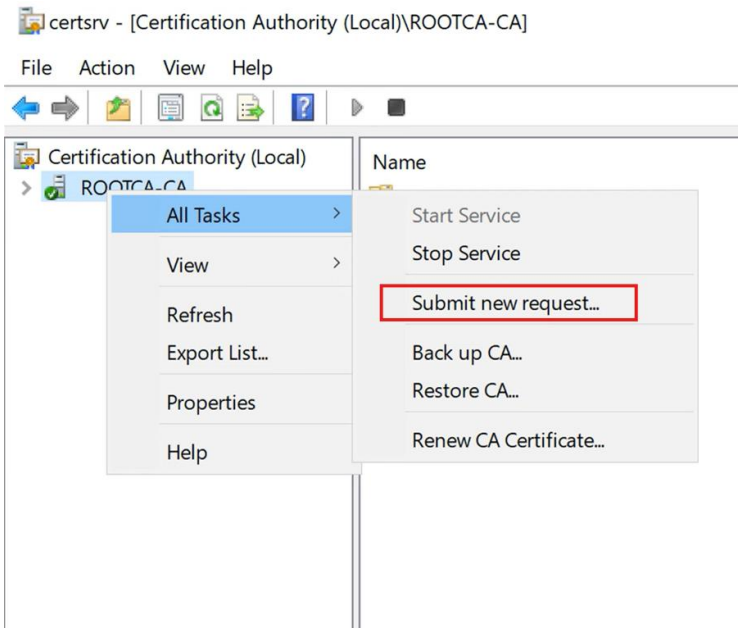



8. **In the pop-up box, do not generate a new private/public key**
9. Save the **.req** file onto a secure media

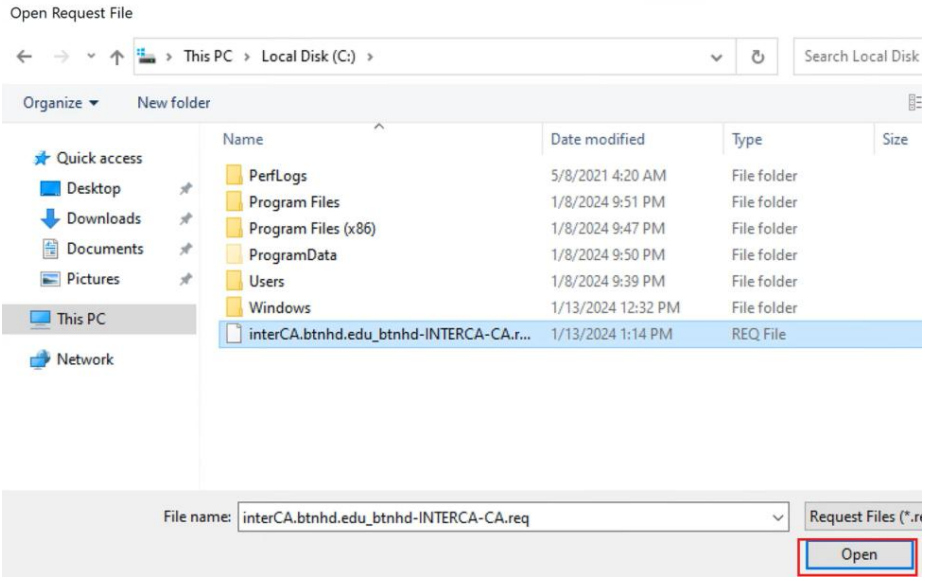
## (4) Signing the Intermediate CSR / Installing the Signed CSR

**On Root CA Server**

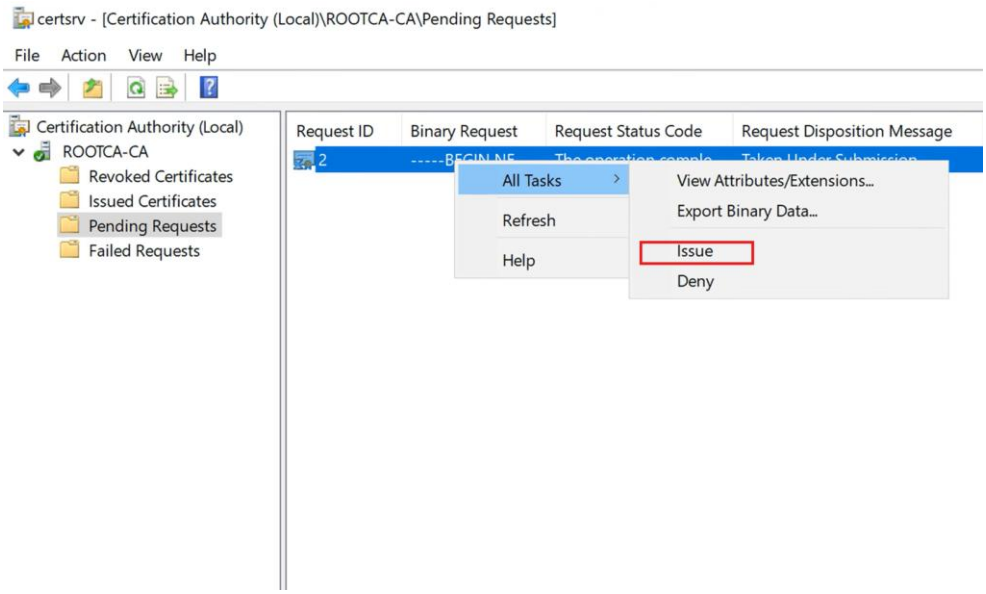
1. Mount secure media containing the **.req** file
2. Copy the latest **.req** file onto **C:**
3. In **Certificate Authority**, right-click the **CA Name** > go to **All Tasks** > click **Submit new request...**



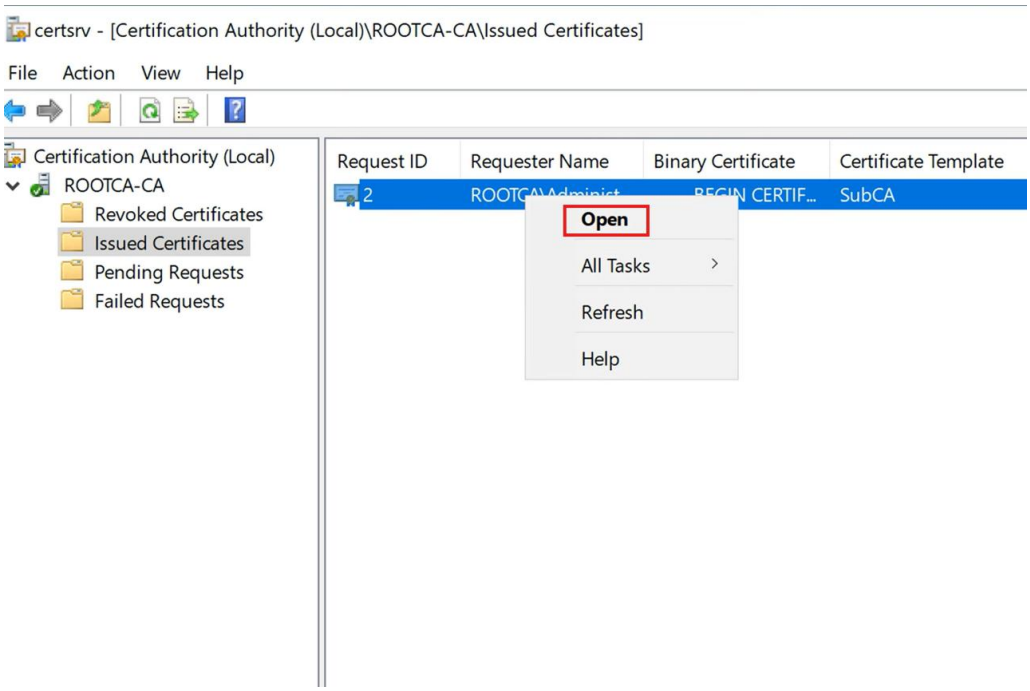

4. In the **Open Request File** window, open the latest imported **.req** file



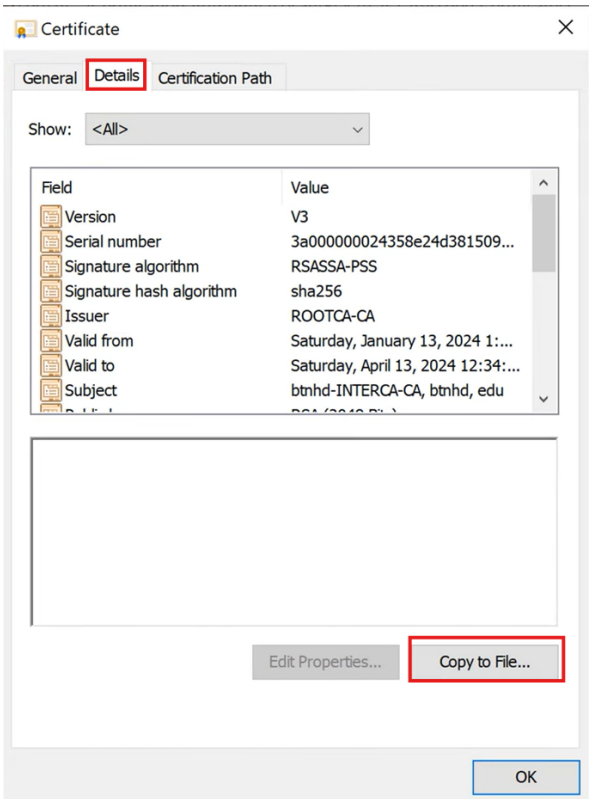
5. In **Certificate Authority**, go to **Pending Requests** > **right-click** the latest request > go to **All Tasks** > and click **Issue**



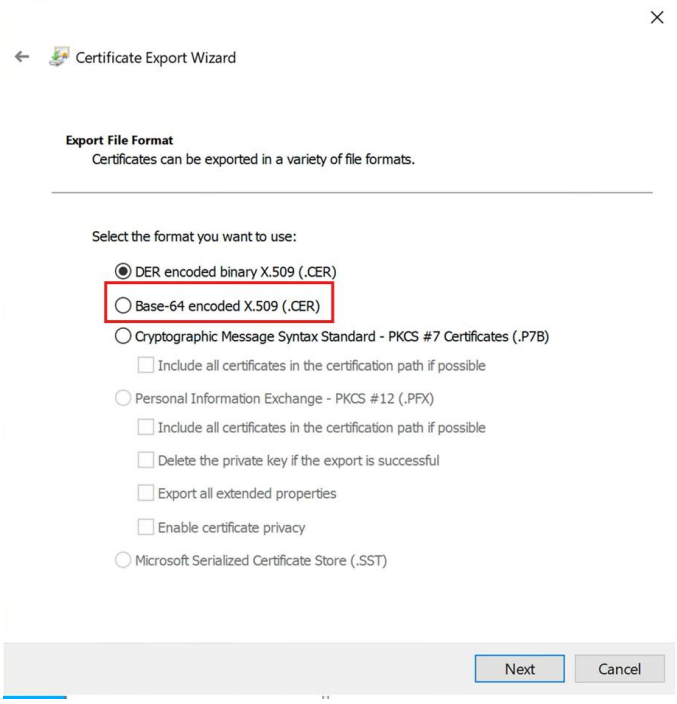

6. Go to **Issued Certificates > right-click** the latest certificate, and click **Open**



7. In the **Certificate** window, go to **Details** tab and **click Copy to File…**



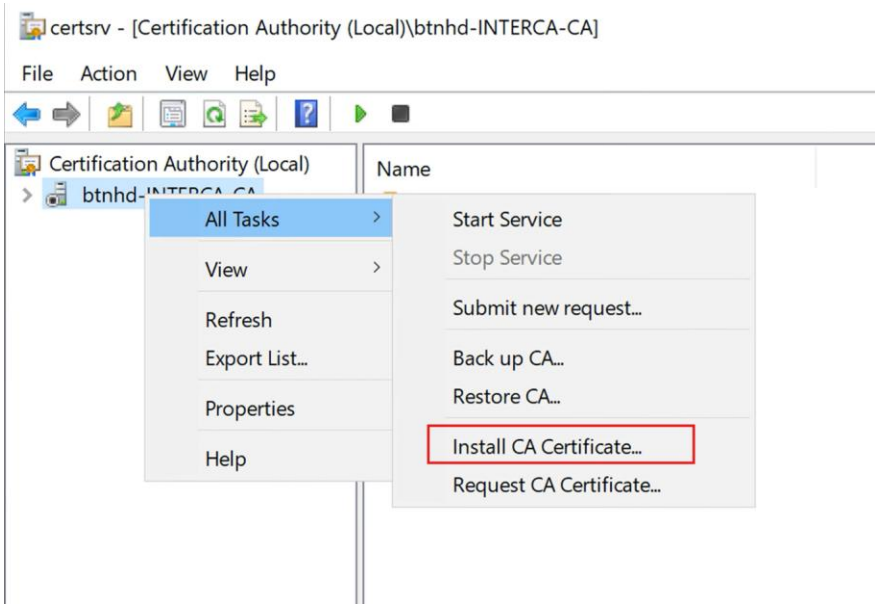

8. In **Certificate Export Wizard**, set the format to be **Base-64 encoded X.509 (.CER)** (this format should be readable for all SCADA device)



9. Continue with defaults, **Save** file name as **[domain_IntermediateCAName_CA].crt**
10. Transfer the latest InterCA **.crt** file onto secure media

**On Intermediate CA Server**

5. Mount secure media containing latest InterCA **.cer/.crt** file
6. In **Certificate Authority**, right-click the **CA Name** > go to **All Tasks** > select **Install CA Certificate**




7. Open the latest Intermediate CA **.cer/.crt** file
8. In **C:\inetpub\wwwroot\CertEnroll**,
    a. Replace the old Intermediate CA Name **.cer/.crt** file with the latest (name must be the same)

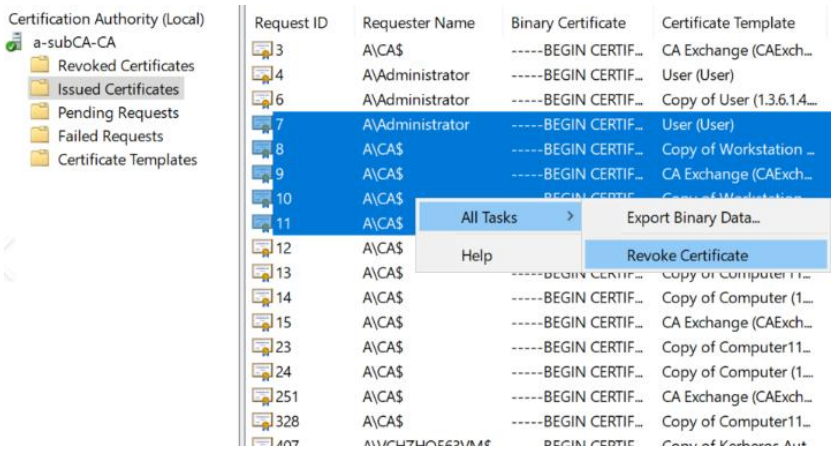
**Restart the Certificate Services service**

# Revoking Certificates (Ad-Hoc Basis)

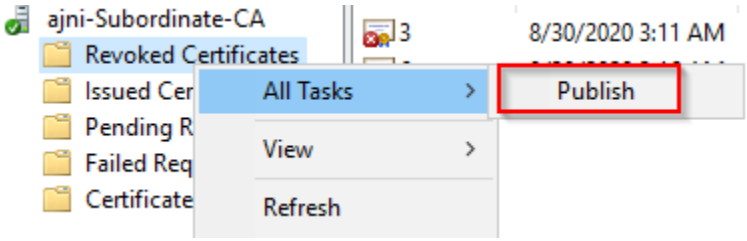
## Revoking A Child Device's Certificate

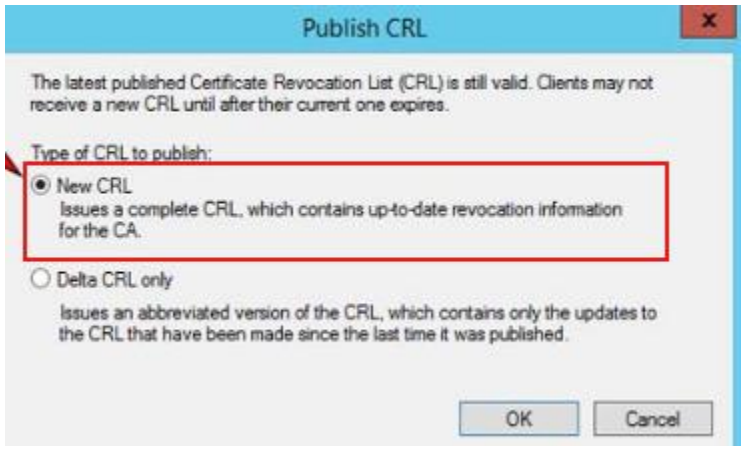
**On Intermediate CA Server**

1. In **Certificate Authority**, go to **Issued Certificates** > right-click the **target certificate** > select **Revoke Certificate**




2. Right-click **Revoked Certificates** > **All Tasks** > **Publish** and choose **New CRL**




**New .crl file in C:\Windows\System32\CertSrv\CertEnroll**

3. In **C:\inetpub\wwwroot\CertEnroll**, replace the old **.crl** file with latest one (**Names must match)**

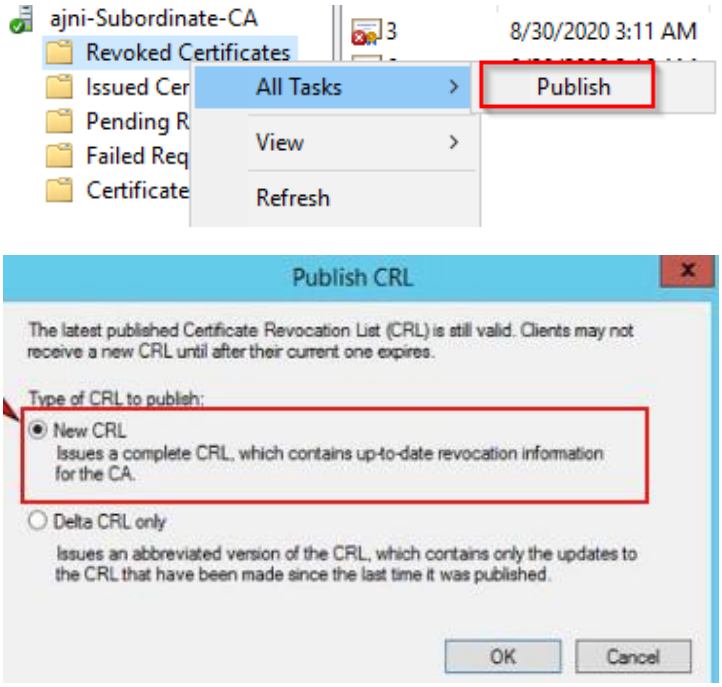
**Windows: As of 4/30/2026, our Intermediate CA is set to auto-update CRL every X days. If wish to immediately update device's CRL you must run "certutil -urlcache * delete" on every device**

**Non-Windows: Some allow installation of CRL, if so, you must manually install on all capable devices**

## Revoking Intermediate CA's Certificate

**Since the Intermediate CA server also hosted the IIS webserver, the whole Intermediate CA component needs to be rebuilt**

1. Power and logon to the Offline Root CA server
2. In **Certificate Authority**, go to **Issued Certificates** > right-click the **Intermediate CA certificate** > **All Tasks** > select **Revoke Certificate**
3. Right-click **Revoked Certificates** > **All Tasks** > **Publish** and choose **New CRL**




**New .crl file in C:\Windows\System32\CertSrv\CertEnroll**

4. Follow **SOP - Step 2 Online Intermediate CA Creation**
    a. **Remember to import the new Root CRL**
    b. **Must have same FQDN**
5. Follow **SOP - Step 3 Domain CA Distribution with Auto Enrollment**
    a. For section *Importing the Root CA and Intermediate CA in the GPO*, **only need to import the Intermediate CA certificate**
    b. Each
6. Follow **SOP - Step 4 Non-Windows Certificate Template**
7. To recertify devices as well as push new CRL,
    a. **For Windows,** previously "certified" devices need to run the following:
        i. **gpupdate /force** (to get new certs)
        ii. **certutil -urlcache * delete** (to forget old CA)
    b. **For Non-Windows,** previously "certified" devices need a new manual certification

## Disaster Recovery During Root Compromise

**If server is breached or private key is stolen, entire PKI environment must be rebuilt**

1. Power down / wipe the Root and/or Intermediate CA servers
2. Follow **SOP – Step 1 Offline Root CA Creation**
3. Follow **SOP – Step 2 Online Intermediate CA Creation**
4. On **Domain Controller,** right-click our "PKI" GPO > select **Edit**
    a. Go to **Computer Configuration > Policies > Windows Settings > Security Settings > Public Key Policies**
    b. For **Trusted Root Certification Authorities**, remove the old Root CA certificate
    c. For **Intermediate Certification Authorities**, remove the old Intermediate CA certificate
5. Follow **SOP – Step 3 Domain CA Distribution with Auto Enrollment**
    a. Skip section *Creating the GPO in Domain Controller*
    b. Skip section *Enabling Auto-Enrollment Policy in the GPO*
6. Follow **SOP - Step 4 Non-Windows Certificate Template**
7. Follow **SOP - Step 5 Non-Windows CA Distribution**